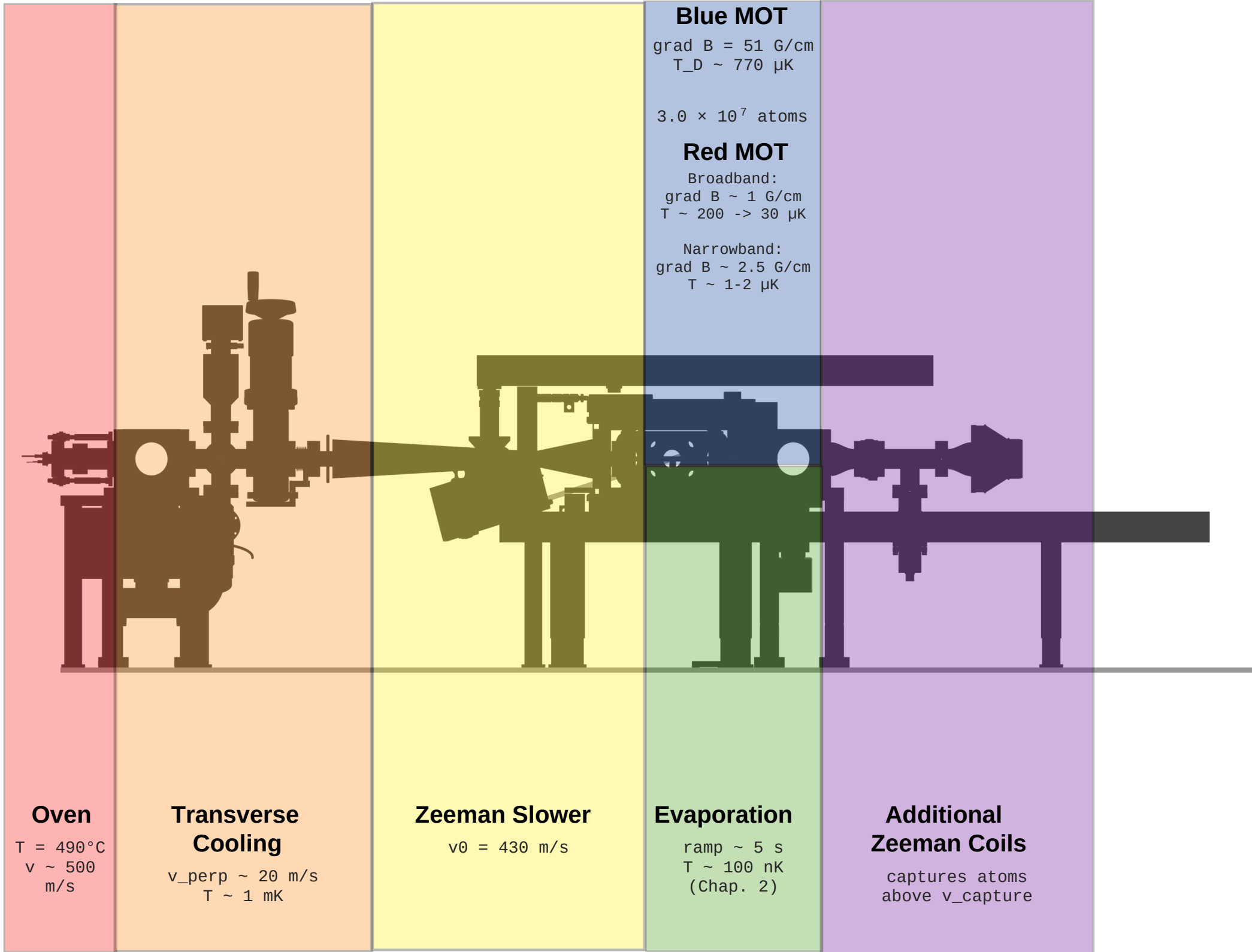




Oven

$T = 490^{\circ}\text{C}$
 $v \sim 500 \text{ m/s}$

Transverse Cooling

$v_{\text{perp}} \sim 20 \text{ m/s}$
 $T \sim 1 \text{ mK}$

Zeeman Slower

$v_0 = 430 \text{ m/s}$

Evaporation

ramp $\sim 5 \text{ s}$
 $T \sim 100 \text{ nK}$
(Chap. 2)

Additional Zeeman Coils

captures atoms above v_{capture}

Blue MOT

grad $B = 51 \text{ G/cm}$
 $T_D \sim 770 \text{ } \mu\text{K}$

$3.0 \times 10^7 \text{ atoms}$

Red MOT

Broadband:
grad $B \sim 1 \text{ G/cm}$
 $T \sim 200 \rightarrow 30 \text{ } \mu\text{K}$

Narrowband:
grad $B \sim 2.5 \text{ G/cm}$
 $T \sim 1\text{-}2 \text{ } \mu\text{K}$